

The Time Value of Money

The time value of money is one of the most fundamental concepts in finance. It states that a sum of money available today is worth more than the same sum in the future, because money available now can be invested to earn a return. This single idea underpins the valuation of loans, investments, bonds, and businesses.

Future Value and Present Value

Future value (FV) measures what a present sum will grow to after earning interest. With compound interest, the future value of a present amount (PV) invested for n periods at an interest rate r per period is $FV = PV \times (1 + r)^n$. Present value runs the calculation in reverse: it is the value today of a future cash flow, computed as $PV = FV$ divided by $(1 + r)^n$. The interest rate used to translate future cash flows into present value is called the discount rate.

Simple vs Compound Interest

Simple interest is calculated only on the original principal, so it grows linearly. Compound interest is calculated on the principal plus all previously accumulated interest, so it grows exponentially—interest earns interest. The more frequently interest is compounded (annually, quarterly, monthly, or daily), the greater the final amount. A useful shortcut is the Rule of 72: dividing 72 by the annual percentage interest rate gives the approximate number of years for an investment to double. For example, at 8 percent, money doubles in roughly nine years.

Annuities

An annuity is a series of equal cash flows made at regular intervals. In an ordinary annuity, payments occur at the end of each period; in an annuity due, payments occur at the beginning of each period. Because each payment in an annuity due is received one period earlier, an annuity due always has a higher present value than an otherwise identical ordinary annuity.

Net Present Value

Net present value (NPV) applies these principles to investment decisions. It discounts all of a project's expected future cash flows back to the present using the discount rate and subtracts the initial investment. A positive NPV indicates that a project is expected to add value and is generally worth pursuing, while a negative NPV suggests it would destroy value.